

Quantitative Transport Through Shifted Collatz Merge Chains

Collatz proof project — September 5, 2026

Write $T(n) = n/2$ for even n , $T(n) = (3n + 1)/2$ for odd n , and $C_i(n) = 3^{j_i(n)}/2^i$, where j_i counts odd steps. Full noncontraction means $C_i(n) \geq 1$ for every i . Shifting such an orbit does not in general preserve that property. An explicit coefficient deficit makes the shifted orbit usable in further conditional descent.

Theorem 1 (Deficit transport). *Suppose $D \geq 1$, $C_i(y) \geq 1/D$ for all i , and $T^s(x) = T^t(y)$. Assume $C_i(x) \geq 1$ for $0 \leq i \leq s$. Put $\rho = C_s(x)/C_t(y) > 0$. Then*

$$C_i(x) \geq \frac{1}{D'} \quad \text{for all } i, \quad D' = \max(1, D/\rho).$$

If N is fully noncontracting, $y = T^k(N)$ satisfies the input bound with $D = C_k(N)$. Along a chain with boosts $\rho_r \geq 1$, the final deficit is $\max(1, C_k(N)/\prod_r \rho_r)$.

Proof. For $v \geq 0$ the shared endpoint gives $C_{s+v}(x) = \rho C_{t+v}(y) \geq \rho/D$; earlier times use the prefix hypothesis. The shift formula is $C_i(T^k(N)) = C_{k+i}(N)/C_k(N) \geq 1/C_k(N)$. The chain formula follows by induction, using $\rho_r \geq 1$. Lean represents a floor a/b by the natural inequalities $a2^i \leq b3^{j_i(n)}$. Its merge theorem accepts $a' \leq b'$ and $a'b2^s3^{j_t(y)} \leq b'a'2^t3^{j_s(x)}$, with $b > 0$. Thus no numerical approximation enters the formal transport. $\square$

Theorem 2 (An infinite conditional descent family). *For every natural Q , if $N = 447 + 549755813888Q$ is fully noncontracting, then so is $M = 307 + 376572715308Q$, and $0 < M < N$.*

Proof. The following paths and odd counts hold for every Q , by binary-cylinder transport. Each replacement has a one-step odd prefix.

seed	value	meeting identity
y	$767 + 940369969152Q$	$T^{23}(N) = y, \quad j_{23}(N) = 15$
x_1	$511 + 626913312768Q$	$T(x_1) = y$
x_2	$461 + 564859072962Q$	$T(x_2) = T^{17}(x_1), \quad j_{17}(x_1) = 11$
M	$307 + 376572715308Q$	$T(M) = x_2$

The boosts are $3/2$, $65536/59049$, and $3/2$. Successive deficits are

$$\frac{14348907}{8388608}, \quad \frac{4782969}{4194304}, \quad \frac{282429536481}{274877906944}, \quad 1.$$

Apply the transport theorem three times. Finally, $N - M = 140 + 173183098580Q > 0$. All statements in this proof, including the all-quotient transport, are kernel checked in `Collatz.MergeDeficit`. $\square$

Finite evidence and limits. Using the existing 360 certified rules, an exact rational search revisits 2,336 canonical seeds below 2^{18} unresolved by the earlier bounded merge search. At shifts $0 \leq k \leq 64$, stopping when $C_k(N) < 1$, it finds chains for 1,012 seeds; 1,216 others reach coefficient contraction first, and 108 reach the horizon. No state cap is reached. Among the 1,012, exactly 182 have no single-rule witness anywhere in this same range; 101 chain witnesses use shifts at most 18. These are Python census results, not kernel-certified completeness claims. Independent orbit replay checks every reported chain. A cap would be reported as incomplete, separately from exhausting a reachable graph.

Verification and remaining obligation. Five public declarations in `Collatz.MergeDeficit` enter the axiom audit. The search, saved witnesses and independent tests are under `analysis/merge_chain_search*`. A successful chain with final seed below N excludes N from being the least positive noncontracting seed. No theorem forces such a chain for every candidate. This work does not prove nondivergence or exclude nontrivial cycles, and makes no mathematical priority claim.